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### Surgical constructs with collapsing suture loop and methods for securing tissue

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#### Abstract

Surgical constructs and methods are provided for securing soft tissue to bone. One exemplary embodiment of a construct is formed from a suture filament and includes two terminal ends of filament and an intermediate portion disposed along at least a portion of a length extending between the terminal ends. The construct can have a first terminal end that is the first terminal end of the filament, and a second terminal end that includes a loop. The loop can be formed by disposing the second terminal end of the filament within a volume of a portion of the intermediate portion of the filament. In some disclosed methods, both terminal ends of the filament can be passed through tissue when performing soft tissue repairs. Various other embodiments of constructs and methods are provided, including constructs having two or more filaments associated with an anchor and methods of using such constructs.

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## **Background/Summary**

PRIORITY (1) The present application is a continuation of and claims priority to U.S. patent application Ser. No. 16/826,432, filed Mar. 23, 2020, and entitled “Surgical Constructs with Collapsing Suture Loop and Methods for Securing Tissue,” which is a continuation of and claims priority to U.S. patent application Ser. No. 15/648,068, filed Jul. 12, 2017, and entitled “Surgical Constructs with Collapsing Suture Loop and Methods for Securing Tissue,” and which issued as U.S. Pat. No. 10,631,848 on Apr. 28, 2020, which is a divisional of and claims priority to U.S. patent application Ser. No. 14/145,486, filed Dec. 31, 2013, and entitled “Surgical Constructs with Collapsing Suture Loop and Methods for Securing Tissue,” and which issued as U.S. Pat. No. 9,737,293 on Aug. 22, 2017, which claims priority to and the benefit of U.S. Provisional

## FIELD

(1) The present disclosure relates to surgical constructs and methods for securing soft tissue to bone, and more particularly relates to surgical constructs having a collapsing suture loop.

## BACKGROUND

(2) A common injury, especially among athletes and people of advancing age, is the complete or partial detachment of tendons, ligaments, or other soft tissues from bone. Tissue detachment may occur during a fall, by overexertion, or for a variety of other reasons. Surgical intervention is often needed, particularly when tissue is completely detached from its associated bone. Currently available devices for tissue attachment include screws, staples, suture anchors, and tacks. Currently available devices for patients of advancing age can be particularly insufficient due to soft and weak bones leading to inadequate fixation between the anchor and bones and the anchors and filaments with which the anchors are coupled.

(3) Repair constructs made from one or more surgical filaments are typically used in soft tissue repair procedures to secure the tissue in a desired location. The repair constructs are typically disposed through one or more portions of the tissue to be repaired, which can cause trauma to the tissue, and are often coupled to anchors disposed in bone to which the tissue is to be approximated. While devices and techniques have been developed to help minimize trauma associated with passing repair constructs through tissue, there is still room for further improvement. For example, some repair constructs may include a sleeve disposed around at least a portion of the limbs of filament of the construct. The sleeve can assist in minimizing trauma to tissue, and also in managing the limbs of suture while the construct is being disposed through tissue. However, the sleeve still adds extra size above and beyond the thickness of the suture, and thus can be a source for added trauma to the tissue. Sleeves also add further costs to the constructs.

(4) In other instances, sutures used in repairs can have portions that have a thicker profile, for instance because they have extra loops, knots, or other configurations formed therein as part of the construct. Thicker profiles can, not surprisingly, provide additional sources of trauma for the tissue as these portions of the construct pass through the tissue. As a result, procedures can often include extra steps to avoid passing portions of the construct that have thicker profiles through tissue. For example, in some instances, a shuttle suture is used to pull the construct through tissue so that an end with a thicker profile does not have to be passed through tissue. The use of additional components, such as a shuttle suture, however, can be cumbersome. Further, to generally avoid trauma to tissue, some procedures may only involving passing a construct through tissue one time. This approach, however, can lead to unsecure tissue attachments due to the footprint of the construct holding the tissue not being large enough.

(5) Still further, there remains a desire to minimize the number of knots used in conjunction with the repair construct when performing soft tissue repair procedures. A variety of different knots, such as sliding knots, can be used to help draw and secure soft tissue with respect to bone. Although the tying of knots at a surgical site is common, in some instances knots can have a tendency to slip, which in turn can cause a loss of tension between the tissue and bone. This drawback is sometimes referred to as a loss of "loop security." In addition to this "loop security" issue, conventional knots typically have an overall size that can be obstructive or intrusive, especially in tight joints, which may damage cartilage or other tissue by abrasion with the knot.

(6) It is therefore desirable to provide repair constructs and methods that reduce the amount of trauma associated with using repair constructs while maintaining or improving the holding strength such constructs and methods can provide. It is also desirable to provide surgical repair methods that reduce the number of steps performed without losing the integrity of the repair performed. Further,

it is desirable to provide constructs and methods for use in soft tissue repair that minimize or eliminate the number and size of knots to be tied by a surgeon, particularly during arthroscopic repair procedures.

## SUMMARY

(7) Surgical constructs and methods are generally provided for securing soft tissue to bone. In one exemplary embodiment, the surgical construct is formed from a suture filament and includes a first terminal end of the filament, a second terminal end of the filament, and an intermediate portion of the filament disposed along at least a portion of a length extending between the first and second terminal ends. The construct can further include a coaxial region formed by the first terminal end being disposed within a volume of a portion of the intermediate portion, and a continuous, closed loop extending from a first side of the coaxial region. The loop can have a first end that is directly adjacent to the first side of the coaxial region, and a second, opposed end formed by a fold in the suture filament. The second, opposed end of the loop can be a first terminal end of the construct, while the second terminal of the filament can extend from a second side of the coaxial region and be a second terminal end of the construct.

(8) A length of the loop of the construct can be configured to be adjusted by moving the first terminal end of the filament with respect to the coaxial region. The construct itself can have a variety of configurations and sizes. By way of non-limiting examples, in some embodiments a length of the coaxial region can be at least about 15.24 centimeters. Likewise, in some embodiments a length of the loop can be at least about 20.32 centimeters. A pick count of the coaxial region can be in a range of about 30 picks per 2.54 centimeters to about 60 picks per 2.54 centimeters. In one exemplary embodiment the pick count of the coaxial region is approximately 40 picks per 2.54 centimeters. In some embodiments, the construct can include an anchor having a filament engagement feature at a distal end thereof. The suture filament can engage the filament engagement feature such that the first terminal end of the construct extends from one side of the filament engagement feature and the second terminal end of the construct extends from an opposite side of the filament engagement feature.

(9) One exemplary embodiment of a surgical repair method includes inserting an anchor in bone and in proximity to detached soft tissue. The anchor can have a surgical filament associated therewith, with the filament having a first portion with a loop formed therein, a second portion that includes a terminal end, and a coaxial region formed between the first and second portions by disposing a portion of the filament into its own volume. The method can include passing the terminal end of the second portion of the filament through a portion of the detached soft tissue, forming a snare in the loop of the first portion of the filament after the anchor has been inserted in bone in proximity to detached soft tissue, and passing the terminal end of the second portion through the snare. The snare can be collapsed to engage the soft tissue and advanced distally to bring the tissue into proximity with the bone. The coaxial region can be deconstructed by removing the portion of the filament from the volume of the filament in which it was disposed.

(10) Another step that can be provided as part of the method includes passing the loop through a portion of the detached soft tissue before forming the snare in the loop. Further, in some instances the method can include tying at least one locking knot with the filament at a location that is proximate to the collapsed snare to secure a location of the filament with respect to the tissue.

(11) In some embodiments the anchor can have a second surgical filament associated with it. The second filament can include a first portion with a loop formed therefrom, a second portion that includes a terminal end, and a coaxial region formed between the first and second portions by disposing a portion of the filament into its own volume. In such an embodiment, the method can further include passing the terminal end of the second portion of the second filament through a portion of the detached soft tissue and passing the loop of the second filament through a portion of the detached soft tissue. After the loop has been passed through the tissue, a snare can be formed in that loop. The terminal end of the second portion of the second filament can be passed through the

snare of the second filament, the snare can be collapsed to engage the soft tissue, and then the collapsed snare can be advanced distally to bring the tissue into proximity with the bone. The coaxial region of the second filament can be deconstructed by removing the portion of the second filament from the volume of the second filament in which it was disposed. Deconstructing the coaxial regions of the first and second filaments can result in a first limb and a second limb extending from their respective collapsed snares. The first and second limbs of both filaments can then be attached to a second anchor, and tension can be applied to the limbs to secure a location of the same with respect to the second anchor. For example, tension can be applied by inserting the second anchor into bone with the first and second limbs being disposed between an outer wall of the anchor and the bone. In some embodiments, the locations of the first and second limbs of the filaments with respect to the second anchor can be secured without tying a knot in either of the first or second filaments.

(12) In another embodiment in which a second surgical filament is provided, the filament can have a first portion with a snare formed therein, a second portion that includes a terminal end, and a coaxial region formed between the first and second portions by disposing a portion of the filament into its own volume. The snare in the first portion can be pre-existing, or alternatively, it can be formed in the construct at some point during the procedure. In addition to using the first filament as described earlier, the method can further include passing the terminal end of the second portion of the second filament through a portion of the detached soft tissue and passing the first portion of the second filament through an opening extending through a length of the anchor. The terminal end of the second filament can be passed through the snare of the second filament, the snare can be collapsed to engage the soft tissue, and then the collapsed snare can be advanced distally to bring the tissue into proximity with the bone. As a result, the first filament can be disposed between the collapsed snare of the second filament and the soft tissue.

(13) Another exemplary embodiment of a surgical repair method includes inserting an anchor in bone in proximity to detached soft tissue. The anchor can have a surgical construct associated therewith, with the construct having a first terminal end and a second terminal end. The method can include passing the first terminal end through a portion of the detached soft tissue, passing the second terminal end through a portion of the detached soft tissue, and then subsequently forming a Lark's Head cinch loop in the second terminal end of the surgical construct such that the Lark's Head cinch loop defines a receiving opening. The first terminal end of the construct can be passed through the receiving opening, the receiving opening can be collapsed, and the collapsed Lark's Head cinch loop can be advanced distally to bring the tissue into proximity with the bone.

(14) The surgical construct can be formed from a surgical filament having a first terminal end, a second terminal end, and an intermediate portion extending therebetween, with the first terminal end of the construct being the first terminal end of the filament, and the second terminal end of the construct having a loop formed by the second terminal end of the filament being disposed in a portion of the intermediate portion of the filament. The method can further include removing the second terminal end of the filament from the intermediate portion of the filament after collapsing the receiving opening of the Lark's Head cinch loop, and tying at least one knot with the filament at a location that is proximate to the collapsed Lark's Head cinch loop to secure a location of the filament with respect to the tissue.

(15) In some embodiments, the anchor can have a second surgical construct associated therewith. The second construct can be formed from a second surgical filament having a first terminal end, a second terminal end, and an intermediate portion extending therebetween. A first terminal end of the second construct can be the first terminal end of the second filament, and a second terminal end of the second construct can have a loop formed by the second terminal end of the second filament being disposed in a portion of the intermediate portion of the second filament. In addition to using the first construct as described earlier, the method can further include passing the terminal end of the second construct through a portion of the detached soft tissue and passing the second terminal

end of the second construct through a portion of the detached soft tissue. After the first and second terminal ends of the second construct have been passed through portions of the detached tissue, a Lark's Head cinch loop can be formed in the second terminal end of the second construct, with the knot defining a receiving opening. The first terminal end of the second construct can be passed through the receiving opening of the second construct, the opening can be collapsed, and then the collapsed Lark's Head cinch loop can be advanced distally to bring the tissue into proximity with the bone. After collapsing the receiving opening of the Lark's Head cinch loop, the method can further include removing the second terminal end of the first filament from the intermediate portion of the first filament and removing the second terminal end of the second filament from the intermediate portion of the second filament to provide first and second limbs of each of the first and second filaments extending from the respective collapsed Lark's Head cinch loops. The first and second limbs of both filaments can then be attached to a second anchor, and tension can be applied to the limbs to secure a location of the same with respect to the second anchor. For example, tension can be applied by inserting the second anchor into bone with the first and second limbs being disposed between an outer wall of the anchor and the bone. In some embodiments, the locations of the first and second limbs of the filaments with respect to the second anchor can be secured without tying a knot in either of the first or second filaments.

(16) In another embodiment in which a second surgical construct is associated with the anchor, the construct can include a first terminal end and a second terminal end, the second terminal end having a snare formed therein. The snare in the second terminal end can be pre-existing, or alternatively, it can be formed in the construct at some point during the procedure. In addition to using the first construct as described earlier, the method can further include passing the first terminal end of the second construct through a portion of the detached tissue and passing the second terminal end of the second construct through an opening extending through a length of the anchor. The first terminal end of the second construct can be passed through the snare of the second construct, the snare can be collapsed, and then the collapsed snare can be advanced distally to bring the tissue into proximity with the bone. As a result, the first construct can be disposed between the collapsed snare of the second construct and the soft tissue.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) This invention will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1A is a schematic view of one exemplary embodiment of a surgical repair construct formed from a suture filament;

(3) FIG. 1B is a detail view of a coaxial region of the surgical repair construct of FIG. 1A;

(4) FIG. 2 is a schematic view of the surgical repair construct of FIG. 1A coupled to a suture anchor;

(5) FIG. 3A is a schematic view of another exemplary embodiment of a surgical repair construct, the construct being formed from two separate suture filaments, each filament having the configuration of the filament of the repair construct of FIG. 1A, and each filament being coupled to a suture anchor;

(6) FIG. 3B is a schematic view of still another exemplary embodiment of a surgical repair construct, the construct being formed from three separate suture filaments, each filament having the configuration of the filament of the repair construct of FIG. 1A, and each filament being coupled to the suture anchor of FIG. 3A;

(7) FIGS. 4A-4N are sequential, schematic, cross-sectional views of one exemplary embodiment for using the surgical repair construct of FIG. 2 to secure tissue to bone;



- (8) FIG. 5A is a schematic, cross-sectional view of an alternative step that can be performed in place of the step illustrated in FIG. 4N;
- (9) FIG. 5B is a schematic, top view of the alternative step of FIG. 5A performed in conjunction with using the construct of FIG. 3A;
- (10) FIGS. 6A-6C are sequential, schematic views of one exemplary embodiment for forming a Lark's head cinch loop as used in the step illustrated in FIG. 4H;
- (11) FIGS. 7A-7F are sequential, schematic, cross-sectional views of one exemplary embodiment for using the surgical repair construct of FIG. 1 to secure tissue to bone, with FIG. 7C illustrating an alternative configuration of the construct to FIG. 7B;
- (12) FIG. 8 is a schematic, top view of another exemplary embodiment for using the surgical repair construct of FIG. 1 to secure tissue to bone;
- (13) FIG. 9 is a schematic, top view of one exemplary embodiment for using the surgical repair construct of FIG. 3A to secure tissue to bone;
- (14) FIG. 10 is a schematic, top view of one exemplary embodiment for using the surgical repair construct of FIG. 2;
- (15) FIG. 11 is a schematic, top view of still another exemplary embodiment for using the surgical repair construct of FIG. 2;
- (16) FIG. 12 is a schematic, top view of yet another exemplary embodiment for using the surgical repair construct of FIG. 2;
- (17) FIG. 13 is a schematic, top view of one exemplary embodiment for using the surgical repair construct of FIG. 3B;
- (18) FIG. 14 is a schematic, top view of one exemplary embodiment for using the surgical repair construct of FIG. 1A in conjunction with the surgical repair construct of FIG. 2; and
- (19) FIG. 15 is a schematic, side view of another exemplary embodiment for using the surgical repair construct of FIG. 3A.

#### DETAILED DESCRIPTION

(20) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed constructs and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such constructs and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape. Sizes and shapes of the constructs, and the components thereof, can depend at least on the anatomy of the subject in which the constructs will be used, the size and shape of components with which the constructs will be used, and the methods and procedures in which the constructs will be used.

(21) The figures provided herein are not necessarily to scale. Further, to the extent arrows are used to describe a direction a component can be tensioned or pulled, these arrows are illustrative and in no way limit the direction the respective component can be tensioned or pulled. A person skilled in the art will recognize other ways and directions for creating the desired tension or movement. Likewise, while in some embodiments movement of one component is described with respect to another, a person skilled in the art will recognize that other movements are possible. Additionally, although terms such as “first” and “second” are used to describe various aspects of a component, e.g., a first end and a second end, such use is not indicative that one component comes before the

other. Use of terms of this nature may be used to distinguish two similar components or features, and often such first and second components can be used interchangeably. Still further, a number of terms may be used throughout the disclosure interchangeably but will be understood by a person skilled in the art. By way of non-limiting example, the terms “suture” and “filament” may be used interchangeably.

(22) Surgical repair constructs and methods for soft tissue repair are generally provided and they generally involve the use of surgical filaments that are configured in a variety of manners to minimize and/or eliminate the tying of knots during a surgical procedure while also minimizing the amount of trauma imparted by the constructs to tissue with which the constructs are used. The constructs described herein provide superior strength for use in a number of different surgical procedures, such as rotator cuff and instability repair procedures and other types of tendon and tissue repair procedures. The designs of the constructs described herein are such that they have a particularly low profile, thereby allowing both terminal ends of a construct to pass through the tissue with minimal trauma to the tissue and to become associated with the tissue without tying knots. The low profile results from inserting one limb of filament into another and eliminating any sort of sleeve, which is often used to assist in shuttling limbs of filament through tissue. The ability to pass both terminal ends of the construct through tissue with minimal trauma to the tissue results from the aforementioned low profile configuration and because the disclosed constructs are generally configured to have one terminal end that is a single filament and a second terminal end that is a loop in which opposed portions of the loop each are formed of a single filament. Passing a single filament, or two single filaments as part of a loop, through tissue results in minimal trauma to the tissue.

(23) As shown by one exemplary embodiment of a surgical repair construct **10** in FIG. 1A, the construct **10** can generally be formed from a single elongate filament **20**, which is used in the formation of three distinct portions of the construct **10**: a loop **12**, a coaxial region **14**, and a tail **16**. The filament **20** includes two terminal ends **20a**, **20b** with an intermediate portion extending therebetween. The first terminal end **20a** can also be a terminal end **10a** of the construct **10**. Thus, as shown, the first terminal end **20a** is part of the tail **16**. The second terminal end **20b**, which is illustrated in FIG. 1B, can be part of the coaxial region **14** formed in an intermediate portion of the construct **10**. More particularly, as shown, the second terminal end **20b** passes through an opening **22** of an outer surface of the intermediate portion of the filament **20** to form a loop closure **24**, and then extends into a volume of the filament **20**. The opening **22** can be manually formed in the filament **20**, for instance by puncturing it, or it can be an opening that forms as part of a braided configuration.

(24) At least a portion of the volume of the filament **20** can be hollow to receive the terminal end **20b**. In some instances, the entire volume of the filament **20** can be hollow. This can be achieved, for example by removing a core from the entire length of the filament **20** using techniques known to those skilled in the art. In other embodiments, only a portion of the core is removed, such as the portion in which the terminal end **20b** will be disposed. Still further, in other embodiments no core may be removed such that the second terminal end **20b** and the core are both disposed within the volume of the filament **20**. Alternatively, or in addition to, in instances in which the filament **20** is a braided filament, the portion of the filament **20** configured to be the volume that receives the terminal end **20b** to form the coaxial region **14** can have a reduced pick count. Reducing the pick count can provide for additional flexibility, and thus additional volume, to receive the terminal end **20b**.

(25) The loop closure **24** can be a self-maintaining junction. As a result, pulling on the tail **16** does not cause the terminal end **20b** to pull out of the volume of the filament **20** in which it is disposed. Rather, pulling on the tail **16** can actually force the volume of the filament **20** to collapse around the terminal end **20b**, thereby providing sufficient friction between the terminal end **20b** and the filament **20** to hold them together. The terminal end **20b**, however, can be removed from the

volume manually at the opening **22** by applying a sufficient amount of force.

(26) Although in the illustrated embodiment the loop closure **24** is formed by inserting the second terminal end **20b** into a volume of the filament **20**, a person skilled in the art will understand other ways by which this junction can be formed without departing from the spirit of the present disclosure. By way of non-limiting examples, the terminal end **20b** can be adhered to the filament **20**, passed from one side of the filament **20** through to the other side, or it can be wrapped around the filament **20** and held in place by one or more features known to those skilled in the art. For instance, a removable pin or flexible member can be passed across the terminal end **20b** and a portion of the filament **20** that serves as the coaxial region **14** to maintain the location of the terminal end **20b** with respect to other portion of the filament **20**. When the pin is removed, the terminal end **20b** can then be moved with respect to the portion of the filament **20** that serves as the coaxial region **14**. Additional disclosures related to such a pin or flexible member are provided in U.S. Patent Application Publication No. 2013/0296931, the content of which is incorporated herein by reference in its entirety.

(27) The second terminal end **20b** is not only a part of the coaxial region, but it also helps define a length  $\alpha$  of the loop **12**. The loop **12** can be a continuous, closed loop configured to have an adjustable length. More particularly, the second terminal end **20b** can be configured to move with respect to the volume in which it is disposed to adjust the length  $\alpha$  of the loop **12**. As shown in FIG. **1A**, the loop **12** is generally defined as having a first end **12a** that is directly adjacent to a second side **14b** of the coaxial region **14** and a second end **12b** that is formed by a fold **20f** in the suture filament **20**, the second end **12b** also doubling as the terminal end **10b** of the construct **10**. The continuous, closed loop **12** can be adjustable such that moving the terminal end **20b** with respect to the volume in which it is disposed can change the length  $\alpha$  of the loop **12**. As shown, applying a force in a direction **A** can decrease a length  $\alpha$  of the loop **12**, and moving the second terminal end **20b** out of the opening **22** can increase the length  $\alpha$  of the loop **12**. As described herein, the loop **12** can have a small profile allowing it to be easily passed through tissue while causing a minimal amount of trauma to the tissue.

(28) A first side **14a** of the coaxial region can have the tail **16** extending therefrom. The tail **16** can extend away from the coaxial region **14** and can be used in conjunction with the loop **12** to help both draw soft tissue towards bone and subsequently maintain the location of the tissue with respect to the bone. As shown in FIG. **1B**, a transition **15** between the second side **14b** and the tail **16** is smooth once the terminal end **20b** is not disposed any further in the volume of the filament **20**. The thickness of the filament **20** becomes reduced due to the fact that no other portion of the filament is disposed in the tail **16**. The smooth nature of the transition **15** makes it easier to pass the filament **20** through tissue.

(29) The construct **10** configuration illustrated in FIGS. **1A** and **1B** provides a number of benefits, at least due in part to the second terminal end **20b** being disposed in a volume of an intermediate portion of the filament **20**. For example, the configuration helps with suture management because by disposing an end of the filament within another portion of the filament, it is one fewer end for which a surgeon needs to account. This is particularly useful in embodiments in which multiple constructs are disposed on a single anchor, as described further below, because each construct has multiple ends for which to account. The disclosed configuration essentially halves the number of filament ends to manage. The configuration also helps prevent filament tangling. The terminal end **20b** cannot tangle with other portions of the filament **20**, or other components being used as part of a procedure, when it is disposed within the volume of the filament **20**. A further benefit that results from the configuration is that it affords surgeons better visibility. There are fewer filaments ends that can be in the surgeon's field of view when the terminal end **20b** is disposed within the volume of the filament **20**. Still further, because no additional components are used to help achieve these benefits, this configuration provides a cheap and easy way to pass filaments through tissue, while also taking up a minimal amount of space.

(30) The filament **20** used to form the construct **10** can be any type and material typically used as filament, including a cannulated filament, a braided filament, and a mono filament. The type and strength of the filament can depend, at least in part, on the other components with which the construct is used, such as an anchor, the tissue through which it will be passed or coupled to, and the type of procedure in which it is used. In some embodiments the filament can have a size between about a #5 filament (about 20 gauge to about 21 gauge) and about a #5-0 filament (about 35 gauge to about 38 gauge), and in one exemplary embodiment the filament is a #2 filament (about 22 gauge to about 24 gauge), such as an Orthocord™ filament that is commercially available from DePuy Mitek, Inc., DePuy Mitek Inc., 325 Paramount Drive, Raynham, Massachusetts 02767, or an Ethibond™ filament that is commercially available from Ethicon, Inc., Route 22 West, Somerville, NJ 08876.

(31) The thickness of the filament should provide strength in the connection but at the same time minimize the trauma caused to tissue through which it passes. In some embodiment, different portions of the construct **10** can have different thicknesses, with the thickness being based, at least in part, on the purpose for that portion, the thicknesses of the other portions of the construct, the components or tissue through which that portion may be passed, and the type of procedure in which the construct is used. Orthocord™ suture is approximately fifty-five to sixty-five percent PDS™ polydioxanone, which is bioabsorbable, and the remaining thirty-five to forty-five percent ultra high molecular weight polyethylene, while Ethibond™ suture is primarily high strength polyester. The amount and type of bioabsorbable material, if any, utilized in the filaments of the present disclosure is primarily a matter of surgeon preference for the particular surgical procedure to be performed.

(32) The lengths of the various portions of the construct **10** can likewise depend, at least in part, on the other components with which the construct is used, the tissue through which it will be passed or coupled to, the lengths of the various portions of the construct, and the type of procedure in which the construct is used. As illustrated in FIG. 1A, a length  $\alpha$  of the loop **12** can be in the range of about 12.7 centimeters to about 50.8 centimeters, a length  $\beta$  of the coaxial region **14** can be in the range of about 7.62 centimeters to about 50.8 centimeters, and a length  $\gamma$  of the tail **16** can be in the range of about 12.7 centimeters to about 101.6 centimeters, and in one embodiment the length  $\alpha$  of the loop **12** is approximately 21.59 centimeters, a length  $\beta$  of the coaxial region **14** is approximately 21.59 centimeters, and a length  $\gamma$  of the tail **16** can be about 25.4 centimeters. There is no specific ratio for any of lengths  $\alpha$ ,  $\beta$ , and  $\gamma$ , and thus in other embodiments the lengths  $\alpha$  and  $\beta$  can be different, and one or both of lengths  $\alpha$  and  $\beta$  can be larger than length  $\gamma$ . It can be desirable for the length  $\alpha$  of the loop **12** to be long enough so that it can be disposed outside of a cannula when in use, as described in greater detail below. This makes it easier for the surgeon to work with the construct **10**, and to monitor it to insure no undesirable tangling of portions of the filament **20** occurs. For example, in some embodiments, the length  $\alpha$  of the loop **12** can be at least about 20.32 centimeters, or at least about 21.59 centimeters, or even longer. Similarly, it can be desirable for the length  $\beta$  of the coaxial region **14** to be long enough so that the terminal end **20b** does not fall out of the volume of the filament **20**, and long enough so that when the coaxial region is disassembled, as described in greater detail below, the two limbs that result from the disassociation are long enough to be grasped by a surgeon outside of the cannulas. For example, in some embodiments, the length  $\beta$  of the coaxial region **14** can be at least about 7.62 centimeters, at least about 15.24 centimeters, at least about 20.32 centimeters, or even longer. A length of the filament **20** itself can be in the range of about 38.1 centimeters to about 203.2 centimeters, and in one exemplary embodiment it has a length of about 111.76 centimeters.

(33) In embodiments in which the filament **20** is braided, the pick count of the braid can be adjusted to assist in receiving the terminal end **20b**. For example, the pick count for a portion of the filament **20** configured to receive the terminal end **20b** to form the coaxial region **14** can be approximately in the range of about 30 picks per 2.54 centimeters to about 60 picks per 2.54

centimeters, and in one instance the pick count can be about 40 picks per 2.54 centimeters. A person skilled in the art will recognize that other pick counts can be used depending, at least in part, on the size of the terminal end **20b** to be received, the type of tissue through which the coaxial region **14** will be disposed, and the various desired properties of the overall construct, such as the ease of sliding a filament within the volume of the filament **20**.

(34) As shown in FIG. 2, the construct **10** can be coupled to a suture anchor **80**. The suture anchor **80** can have an internal cannulation **82** extending through a portion thereof and one or more filament engagement features, such as the filament engagement feature **84** disposed at a distal end **80d** of the anchor **80**. In the illustrated embodiment, the loop **12** extends from one side of the anchor **80** and the terminal end **10a**, including at least a portion of the tail **16**, extends from the other side of the anchor **80**. Although in the illustrated embodiment the coaxial region is disposed on the same side as the loop **12**, in other embodiments the coaxial region **14** can be at least partially disposed on the same side as the terminal end **10a**, at least because the construct **10** can generally be configured to slide with respect to the filament engagement feature **84** to move to various locations with respect to the anchor **80**. The anchor can also include one or more external fixation enhancements, such as threads **86**, for engaging bone in which the anchor can be disposed.

(35) One skilled in the art will appreciate that a variety of suture anchor types can be used in conjunction with the constructs provided herein, including both hard and soft anchors, and that the disclosure is not intended to be limited to the designs of anchors provided for herein. Some exemplary embodiments of anchors that can be used in conjunction with the constructs and related teachings provided for herein include a Healix Ti™ anchor, a Healix Advance™ anchor, a Healix Advance™ Knotless anchor, a Versalok™ anchor, and a Gryphon™ anchor, each of which is commercially available from DePuy Mitek, Inc., as well as anchors described in U.S. Patent Application Publication No. 2013/0296934, and U.S. patent application Ser. No. 13/623,429, entitled “Systems, Devices, and Methods for Securing Tissue Using Hard Anchors,” filed Sep. 20, 2012, the content of which is incorporated by reference herein in their entireties.

(36) While FIG. 2 illustrates a single construct associated with the anchor **80**, in other embodiments more than one construct can be loaded onto the anchor **80**. By way of non-limiting examples, FIG. 3A illustrates an embodiment in which constructs **10**, **10'** are double-loaded onto a suture anchor **80'**, and FIG. 3B illustrates an embodiment in which constructs **10**, **10'**, **10''** are triple-loaded onto the suture anchor **80'**. The suture anchor **80'** can include a filament engagement feature (not shown) around which each construct **10**, **10'**, **10''** can be disposed. As a result, the loops **12**, **12'**, and **12''** can be disposed on one side of the anchor **80'** and the terminal ends **10a**, **10a'**, and **10a''** including at least a portion of the tails **16**, **16'**, and **16''** can be disposed on an other side of the anchor **80'**.

(37) Each construct **10**, **10'**, **10''** disposed around the filament engagement feature can be in a touching, side-by-side disposition. In other embodiments one construct can be disposed over a portion of another as they wrap around a portion of the filament engagement feature. The use of multiple constructs with one anchor can increase the resulting footprint of the implant. Further, as the number of constructs associated with the anchor **80'** increases, the benefits of the provided for configurations are magnified. In particular, the suture management benefits are even more pronounced with two and three constructs associated with the anchor **80'** than with one construct. As the number of constructs increase, the size of the constructs can remain the same, or they can decrease to help allow more filaments to be attached to the anchor and used without interfering with the other filaments. By way of non-limiting example, in some exemplary embodiments of a single construct or two constructs being associated with an anchor the filament forming the construct(s) can be a #2 (about 22 gauge to about 24 gauge) Orthocord™ filament, and in some exemplary embodiments of three constructs being associated with an anchor the filament forming the three constructs can be a #0 (about 26 gauge to about 27 gauge) Orthocord™ filament.

(38) The use of multiple constructs can also enhance the type and effectiveness of various tissue repair surgical procedures, including those discussed below and others known to those skilled in

the art. Each construct can have a unique identifier to assist the surgeon in identifying a particular filament during the procedure. Examples of unique identifiers include, but are not limited to, different colors, patterns, or surfaces to provide different tactile feels. Additionally, each construct **10** itself can have unique identifiers associated with each terminal end, thus making it easier for a surgeon to know which end has the loop **12** and which is the tail **16**. Identifying the two different ends of a single construct can be helpful in allowing the surgeon to know which end will serve as a post along which a collapsed Lark's Head cinch loop is distally advanced, as described in greater detail below.

(39) The size of the anchor **80'** can depend on a variety of factors, including, by way of non-limiting example, the type and size of the constructs with which it is used, the bone in which it will be disposed, and the type of procedure in which it will be used, but in some exemplary embodiments it can have an outer diameter in the range of about 3 millimeters to about 6 millimeters, and in one embodiment its outer diameter can be about 4.75 millimeters. Further, although in the illustrated embodiments the constructs **10**, **10'**, and **10''** are associated with an anchor **80'**, other types of implantation devices can also be used in conjunction with the constructs **10**, **10'**, and **10''**. Such devices include, by way of non-limiting example, cortical buttons and other strands of suture filament. Alternatively, the construct **10** can be used independent of an anchor, for instance to help grasp or tie tissue and the like, as described further below and in greater detail in U.S. Patent Application Publication No. 2013/0296931, the content of which was previously incorporated herein by reference.

(40) Exemplary methods for using constructs of the type described herein are now described in further detail. The methods described herein generally relate to attaching soft tissue, such as tendon **105**, to bone, although a person skilled in the art will recognize other types of procedures with which the constructs and the methods related to the same can be used. The embodiment illustrated in FIGS. 4A-4N is performed in a minimally invasive manner, e.g., arthroscopically, through a first cannula **100** and a second cannula **102** that are disposed through a patient's skin **104** using techniques known to those skilled in the art. The use of multiple cannulas can help a surgeon manage filament during a surgical procedure, as well as any instruments used in conjunction with the procedure. In the illustrated embodiment the first cannula **100** is substantially aligned with the location at which the procedure is performed and serves as a working channel. A person skilled in the art will recognize other ways by which the procedures described herein can be performed, including through three or more cannulas, a single cannula, or through no cannula at all. Further, other types of procedures, such as open procedures, can be used in conjunction with the present disclosures.

(41) As shown in FIG. 4A, a bone hole **108** can be formed in bone **106** using techniques known to those skilled in the art. The bone hole **108** can be disposed at a location proximate to the location at which the tendon **105** is to be attached. An anchor **80** and a suture construct **10** can then be passed through the first cannula **100** and implanted in the bone hole **108** using ordinary techniques, such as by using a driver to screw or tap the anchor into place. In the illustrated embodiment of FIG. 4B, the terminal end of the tail **16** extends from one side of the filament engagement feature **84** and out of the cannula **100** and the loop **12** extends from the other side of the filament engagement feature **84** and also out of the cannula **100**. The construct **10** can be coupled to or otherwise associated with the anchor **80** prior to implantation, or it can be thread around the filament engagement feature **84** after the anchor **80** has been implanted in the bone **106**.

(42) As shown in FIG. 4C, a portion of the construct **10** that includes the loop **12** can be moved to the second cannula **102**. This allows for additional visibility in the working channel, and also helps a surgeon manage the two terminal ends **10a**, **10b** of the construct **10** so they do not become tangled and/or interchanged during portions of the procedure. Any technique known to those skilled in the art can be used to effect movement of the tail **16**, and any portion of the filament **20** for that matter. In some exemplary embodiments a tool sometimes referred to as a suture grasper can be

used.

(43) As shown in FIG. 4D, the tail **16** can be passed through the tendon **105** to be attached to bone **106** and can be passed through the second cannula **102**. This results in an unobstructed view in the working channel for additional steps in the procedure. While the tail **16** can be passed through tissue using a number of techniques known to those skilled in the art, in some embodiments a suture passing device such as the EXPRESSEW II flexible suture passer, which is available from DePuy Mitek, LLC, can be passed through the first cannula **100** and operated to pass the tail **16** through the tendon **105**. Furthermore, a tool such as a suture grasper can be used to move the tail **16** into the second cannula **102** after it has been passed through the tendon **105**.

(44) In some procedures, including some described below with respect to other illustrated embodiments, the loop **12** is not passed through tissue. However, one benefit afforded by the configuration of the present construct **10** is that it can be easily passed through tissue while minimizing an amount of trauma resulting from the same. In the present embodiment, the loop **12** is passed through tissue, as shown the tendon **105**. Optionally, as shown in FIG. 4E, the loop **12** can be moved from the second cannula **102** to the first cannula **100** prior to passing the loop **12** through tendon **105**. Doing so can help with suture management and prevent tangling of the filament **20** with itself or other objects disposed at or near the surgical site. The loop **12** can be passed through the tendon **105** and again back to the second cannula **102** using known techniques for passing filament through tissue. The loop **12** can pass through the tendon **105** easily because it is only the two strands of filament **20** that press through the tendon **105**. There are no additional suture management components, knots, or other obstructions associated with the loop **12**. The configuration that results from passing the loop **12** through the tendon **105** and into the second cannula **102** is illustrated in FIG. 4F.

(45) Both the loop **12** and the tail **16** can be returned to the first cannula **100** for use in the working channel, as shown in FIG. 4G. A receiving opening **52** can then be formed in the loop **12**. A number of different techniques can be used to form the receiving opening, but in one exemplary embodiment, which is illustrated in FIG. 4H, a Lark's Head cinching loop **50** is formed from the filament **20** that forms the loop **12**, with the Lark's Head cinching loop **50** being disposed at the distal end **10b** of the construct **10**. FIGS. 6A-6C, which are described in further detail below, describe one exemplary embodiment for forming a Lark's Head cinching loop, although a person skilled in the art will understand numerous techniques that can be used to form a Lark's Head cinching loop. Other types of configurations can also be used to form the receiving opening without departing from the spirit of the present disclosure.

(46) As shown in FIG. 4I, the tail **16** can be passed through the receiving opening **52** to form a collapsing loop **54** that captures the filament engagement feature **84** and the tendon **105**. The Lark's Head cinching loop **50** can then be collapsed or dressed around the portion of the tail **16** disposed therethrough, as shown in FIG. 4J. Further, tension can be applied to the terminal end **10a** of the construct **10** by pulling approximately in a direction C, thereby causing the collapsed Lark's Head cinch loop **50** to slide distally toward the tendon **105** in a zip-line like manner until the collapsed Lark's Head cinch loop **50** is adjacent to the tendon **105**, as shown in FIG. 4K. Alternatively, tension can be applied to the terminal end **10a** of the construct **10** before the Lark's Head cinch loop **50** is dressed and after the Lark's head cinch loop **50** is adjacent to the tendon, or some combination of the two actions can be used, such as partially dressing the Lark's Head cinch loop **50** before zip-lining it toward the tendon **105**. In some embodiments, a knot-pushing tool can be used to assist in advancing the collapsed Lark's Head cinch loop **50** toward the tendon **105**. The collapsed Lark's Head cinch loop **50** provides sufficient holding to maintain tension in the collapsing loop **54**. Further, the collapsed Lark's Head cinch loop **50** provides a wider, low profile configuration that can reduce trauma to the tendon. More particularly, while the implant can be wider because of the two portions of filament extending through the tissue, the distance that the collapsed Lark's Head cinch loop **50** extends away from the tendon **105** is smaller than

configurations previously relied upon for attaching tissue to bone. The configuration can also have a wide profile because portions of the filament **20** are hollow, allowing the filament **20** to become flatter with respect to the tendon **105** in which it is disposed.

(47) Advancing the Lark's Head cinch loop **50** toward the tendon **105** can result in the coaxial region **14** being moved out of the tissue and out of the cannula, as shown in FIG. **4K**. When the coaxial region **14** is out of the tissue and exposed outside of the cannula **100**, it can be easier to disassemble the coaxial region **14** and to prevent portions of the filament **20** from tangling with itself or other components associated therewith. A disassembled coaxial region is illustrated in FIG. **4L**. As described above with respect to FIG. **1B**, the second terminal end **20b** can be untucked or pulled out of the volume of the filament **20** proximate to the opening **22**. The resulting configuration is a pair of limbs **56, 58** extending out of the first cannula **100** that can be used in a variety of ways to complete a tissue attachment procedure.

(48) The compression afforded by the Lark's Head cinch loop **50** can hold the tendon **105** through which the construct **10** is passed at a desired location. Nevertheless, in one instance, illustrated in FIG. **4M**, the limbs **56, 58** can assist in providing further security by being cinched or otherwise tied together to secure the location of the collapsed Lark's Head cinch loop **50**, and thus the tendon **105**, with respect to the bone **106**. In the illustrated embodiment, a half-hitch **60** is formed by the surgeon using the first and second limbs **56, 58**. Tying an initial half-hitch creates a one-way lock that allows for further tightening of the collapsing loop **54**, i.e., by distally advancing the collapsed Lark's Head cinch loop **50**, without expanding the loop, i.e., it prevents the collapsed Lark's Head cinch loop **50** for moving proximally. In some embodiments, a second half-hitch **62** can be formed to lock the location of the first half-hitch **60**, thereby preventing advancement of the collapsed Lark's Head cinch loop **50** in either the distal or proximal directions. As shown in FIG. **4N**, the limbs **56, 58** can be trimmed relatively close to the collapsed Lark's Head cinch loop **50** and associated half hitch(es) **60, 62** formed adjacent thereto, however, care should generally be taken not to cut the limbs **56, 58** too close to the collapsed Lark's Head cinch loop **50** so as not to affect the integrity of the collapsed Lark's Head cinch loop **50** and associated half hitch(es) **60, 62**.

(49) In addition to the aforementioned low profile configuration that results from the described procedure and variations thereof, there are numerous other advantages associated with the construct **10** and its use in surgical procedures. For instance, the resulting strength of the ending configuration is as strong and/or stronger than existing configurations known in the art. This is particularly impressive given that it takes up less space with its low profile.

(50) The strength of the construct **10** was tested by determining a failure load for the construct, i.e., the amount of force at which the collapsing loop **54** expanded 3 millimeters or more. The tested construct was formed from a #2 Orthocord™ filament and was disposed on a 9.5 millimeter dowel pin. The collapsing loop **54** was formed from the filament using techniques described herein. Accordingly, the Lark's Head cinch loop **50** was collapsed to form the collapsing loop **54** and two half-hitches were tied on top of the collapsed cinch loop **50** to lock the location of the cinch loop **50**. The resulting collapsing loop **54** had two limbs of the filament disposed adjacent to each other in a side-by-side configuration on the dowel pin. Load was then progressively applied to the collapsing loop **54** and an amount of expansion of the loop was measured as the amount of load increased. Once the collapsing loop expanded 3 millimeters or more, the amount of load applied was noted as the failure load. During testing of the aforementioned construct configuration, the failure load for the #2 Orthocord™ filament when it was coated with New Vicryl Coating was about 270 Newtons. The failure load for the #2 Orthocord™ filament when it was not coated with an additional material was about 360 Newtons.

(51) Another benefit afforded by the side-by-side configuration is that it helps reduce a tendency of the filament **20** to abrade or cut through the tissue, which can occur more readily when just a single filament extends through the tissue. Still further, the configurations provided for herein are uncomplicated, and to the extent any knot tying is involved, such tying is quick and easy. This



allows surgeons to perform more consistent procedures from patient-to-patient.

(52) In some embodiments, rather than trimming the two tails **56**, **58**, they can be used to associate a second anchor with the first anchor **80** to form a dual-row repair. FIG. 5A represents one such configuration, which is sometimes referred to as a double row spanned repair. As shown, the tails **56**, **58** extend from the one or more half-hitches **60**, **62** and toward a second anchor **80''** disposed in the bone **106** at a location proximate to the first anchor **80**. The second anchor **80''** can be disposed in the bone **106** using techniques known to those skilled in the art, including by forming a bone hole **108''** in which the anchor **80''** can be disposed. In this embodiment, the second anchor **80''** can be a truly knotless anchor, such as a Healix Advance™ Knotless anchor or a Versalok™ anchor because of the use of the two tails **56**, **58** and the anchor **80''** to secure the location of the collapsed Lark's Head cinch loop **50**.

(53) As shown, the limbs **56**, **58** can be placed such that as the anchor **80''** is disposed in the bone hole **108''**, the tails **56**, **58** can be pinched between an outer surface of the anchor **80''** and a wall of the bone hole **108''**. Terminal ends **56t**, **58t** of the limbs **56**, **58** can then be passed through the anchor **80''** and back out a top side **80a''** of the anchor **80''** for subsequent use and/or removal. In the illustrated embodiment, the second anchor **80''** is fully cannulated through a length thereof and does not include a filament engagement feature, while the first anchor **80** is also fully cannulated through its length but does include a filament engagement feature **84**. A person skilled in the art will recognize that a variety of different anchor configurations can be used in conjunction with this surgical technique, and thus the illustrated embodiment of one anchor having a filament engagement feature and a second anchor being cannulated with no filament engagement feature is no way limits the scope of the present disclosure. Further, although in the illustrated embodiment the tails **56**, **58** extend from the one or more half-hitches **60**, **62** formed adjacent to the Lark's Head cinch loop **50**, in other embodiments no half-hitches are formed and the location of the Lark's Head cinch loop **50** is instead secured by applying tension to the tails **56**, **58**. For example, driving the second anchor **80''** into the bone hole **108''** and trapping the tails **56**, **58** between the anchor **80''** and the bone **106** can supply sufficient tension to secure the location of the Lark's Head cinch loop **50**, and thus the tendon **105** itself. Similarly, even a Lark's Head cinch loop can be eliminated such that the tension supplied to the tails **56**, **58** by virtue of being pinched between the anchor **80''** and the bone **106** can be sufficient to maintain a desired location in conjunction with performing a dual-row type repair. A person skilled in the art will recognize a variety of other ways by which the constructs and techniques described herein can be used in conjunction with a dual row type repair.

(54) FIG. 5B illustrates an alternative embodiment of using a lateral row anchor **80''** to help secure the location of the constructs **10**, **10'** with respect to the anchor **80'** of FIG. 3A. This embodiment is sometimes referred to as a double row span procedure. As shown, tails and loops of the two constructs **10**, **10'** can be passed through to a top side of the tissue **105** at locations M, N and P, Q, respectively, as shown. The tails of the first construct **10**, **10'** can then be passed through respective receiving openings **50**, **50'** formed in the loops of the constructs **10**, **10'** the receiving openings **50**, **50'** can be collapsed, and the receiving openings **50**, **50'** can be advanced toward the tissue **105** as shown using techniques previously described above. The coaxial regions of the constructs **10**, **10'** can be disassembled, resulting in limbs **56**, **58** extending from the collapsed receiving opening **50** of the construct **10** and limbs **56'**, **58'** extending from the collapsed receiving opening **50'** of the construct **10'**. The limbs **56**, **58** and **56'**, **58'** can then be trapped between the anchor **80''** and bone **106** in a manner similar to as described with respect to FIG. 5A. In some embodiments, the receiving opening **50**, **50'** can be a Lark's Head cinch loop as described herein. While half-hitches can be used in conjunction with the Lark's Head cinch loop as described herein, the tension to the limbs **56**, **58** and **56'**, **58'** supplied by the anchor **80''** can be sufficient to maintain the location of the constructs **10**, **10'** with respect to the anchor **80'**. Further, in other embodiments, the loop itself can be the receiving openings **50**, **50'** and no half-hitches or other knots can be associated therewith because the location of the constructs **10**, **10'** with respect to the anchor **80'** can be maintained by

the tendon supplied to the limbs **56**, **58** and **56'**, **58'** by the anchor **80''**.

(55) FIGS. **6A-6C** illustrate one exemplary embodiment for forming the Lark's Head cinch loop **50**, which is a process that can be performed in connection with the formation of the receiving opening **52** as illustrated in FIG. **4H**. As shown in FIG. **6A**, the filament **20** can be folded substantially in half at an approximate midpoint **20m** of the filament **20**, forming a first filament limb **51** and a second filament limb **53**. A central portion of the filament, which includes the midpoint **50m**, can be folded toward the first and second limbs **51**, **53** and brought proximate to the first and second limbs **51**, **53**, as shown in FIG. **6B**. This results in the formation of a first sub-loop **55** and a second sub-loop **57**. A size of the sub-loops **55**, **57**, and a length of the remaining portions of the limbs **51**, **53** extending therefrom, can be adjusted as desired. The sub-loops **55**, **57** can then be folded back on themselves, for instance by grasping a portion **51p**, **53p** of the limbs **51**, **53** that are part of the sub-loops **55**, **57** and pulling upward (as shown, "out of the page"). This results in the configuration illustrated in FIG. **6C**, which as shown is the filament **20** having the Lark Head's cinch loop **50** formed therein with remaining portions of the first and second limbs **51**, **53** extending therefrom. The Lark's Head cinch loop **50** defines the collapsible receiving opening **52**, a size of which can be decreased by applying a force in an approximate direction **F** to one or both of the limbs **51**, **53** extending from the loop **50**, or by applying a force in an approximate direction **G** to the opening **52**. Likewise, a size of the opening **52** can be increased by grasping near the midpoint **20m** of the filament **20** to hold the portion where the fold is formed approximately stationary and then applying either a force in the approximate direction **G** to both of the limbs **51**, **53** extending from the cinch loop **50**, or a force in the approximate direction **G** to the opening **52**.

(56) A person skilled in the art will recognize other ways by which a Lark's Head cinch loop can be formed. Similarly, a person skilled in the art will be familiar with other types of cinch loop or knotted configurations, e.g., configurations having sliding knots, that can be formed from suture filaments, and will understand ways in which other techniques can be adapted for use in a manner as the Lark's Head cinch loop is used in the present disclosure. The present disclosure is not limited to use only with a Lark's Head cinch loop.

(57) FIGS. **7A-7F** illustrate another exemplary embodiment for using the construct **10** in a tissue repair procedure. Although not illustrated, a surgical opening can be formed through a patient's skin and one or more cannulas can be passed therethrough to create a surgical repair site in a manner similar to described above with respect to FIGS. **4A-4N** and otherwise known to those skilled in the art. The cannula(s) is not shown for ease of illustration. A bone hole **108** can be formed in bone **106** using techniques known to those skilled in the art, with the bone hole **108** located at a location proximate to the location at which the tissue, which is tendon **105** in the illustrated embodiment, is to be attached. Unlike the configuration described above, in the illustrated embodiment the construct **10** is disposed through tissue prior to being associated with an anchor disposed in the bone **106**. As shown in FIG. **7A**, the tissue is passed through the tissue such that both the tail **16** and the loop **12** extend proximally toward the surgeon for use during the procedure. The construct **10** can be associated with the tendon **105** using a variety of techniques, including by threading the tail **16** through the tissue from a top side **105a** to a bottom side **105b**, and then back again from the bottom side **105b** to the top side **105a**. In an alternative embodiment, both the tail **16** and the loop **12** can be passed through the tendon **105** to achieve the configuration illustrated in FIG. **7A**.

(58) The tail **16** can then be passed through the loop **12**, as shown in FIG. **7B**. Alternatively, as shown in FIG. **7C**, the loop **12** can be formed into a Lark's Head cinch loop **50** using techniques previously described, or others known to those skilled in the art, and then the tail **16** can be passed through the receiving opening **52** defined by the cinch loop **50**. In either configuration, passing the tail **16** through the loop **12** or receiving opening **52** results in the formation of a collapsing loop **54**. As described above, the Lark's Head cinch loop **50** can be collapsed or dressed around the portion of the tail **16** disposed therethrough. Likewise, to the extent that the loop **12** is adjustable, a person skilled in the art would recognize ways by which the loop **12** could be collapsed around the portion

of the tail **16** disposed therethrough. As shown in FIG. 7D, tension can be applied to the terminal end **10a** of the construct **10** by pulling approximately in a direction C, thereby causing the loop **12** (FIG. 7B) or Lark's Head cinch loop **50** (FIG. 7C) to slide distally toward the tendon **105** in a zip-line like manner until the loop **12** or Lark's Head cinch loop **50** is adjacent to the tendon **105**. The collapsed loop **12** and the collapsed Lark's Head cinch loop **50** provide sufficient holding to maintain tension in their respective collapsing loops **54**.

(59) The coaxial region **14** of the construct **10** can subsequently be disassembled, as shown in FIG. 7E. More specifically, a second terminal end of the filament **20** can be untucked or pulled out of the volume of the filament **20** proximate to an opening of the filament **20**, as described above with respect to FIG. 1B. The resulting configuration is a pair of limbs **56**, **58** that can be used in a variety of ways to complete a tissue attachment procedure. By way of non-limiting example, FIG. 7F illustrates the limbs **56**, **58** extending to an anchor **80'''** disposed in the bone hole **108** in a manner as described previously with respect to FIG. 5. In an alternative embodiment, illustrated in FIG. 8, the limb **56** can extend from the collapsed Lark's Head cinch loop **50** associated with the anchor **80'''** (not shown) located in a medial or primary fixation row to an anchor **80c** located in a lateral row, while the limb **58** can extend from the collapsed Lark's Head cinch loop **50** to an anchor **80d** also located in the lateral row. Tension can be applied to the limbs **56**, **58** by the respective anchors **80c**, **80d**, or using other techniques known to those skilled in the art, thereby maintain a location of the collapsed receiving opening with respect to the anchor **80'''**. While as described with respect to FIG. 8 the Lark's Head cinch loop **50** is associated with an anchor, in other embodiments the collapsed Lark's Head cinch loop **50** can be associated with the tissue **105** on its own, i.e., without an anchor, using techniques disclosed herein or otherwise known to those skilled in the art.

(60) FIGS. 9-15 illustrate some further, non-limiting types of tissue securing procedures that can be performed using the constructs and related disclosures provided for herein. The constructs can lead to procedures that are faster, easier, and more durable than existing tissue repair procedures.

(61) FIG. 9 illustrates one exemplary embodiment of a single row repair. As shown, two anchors **80'** are double-loaded such that they both include two constructs **10**, **10'** coupled thereto. The anchors **80'** are disposed in bone **106** using techniques known to those skilled in the art. The loop and tail of one of the constructs **10** associated with each anchor **80'** can be passed through to a top side of the tissue **105** at locations M.sub.1 and N.sub.1 as shown. The tail can then be passed through a receiving opening **52** formed in the loop, the receiving opening **52** can be collapsed, and the collapsed receiving opening **52** can be advanced toward the tissue **105** using techniques described herein. Further, the tail of the other construct **10'** can be passed through to a top side of the tissue **105** at a location P.sub.1 as shown, for engagement with the loop associated with the anchor **80'**. The tail can then be passed through a receiving opening **52'** formed in the loop, the receiving opening **52'** can be collapsed, and the collapsed receiving opening **52'** can be advanced toward the tissue **105** as shown using techniques previously described above. The location P.sub.1 is such that when the tail and loop are coupled together via the receiving opening **52'**, the combination of the tail and loop of the first construct **80** is disposed between the tissue **105** and the combination of the tail and loop of the second construct **10'**. As a result, the combination of the tail and the loop of the first construct **10** can serve as a rip stop or cruciate stitch to help prevent tissue tearing by the other construct **10'**.

(62) FIG. 10 illustrates one exemplary embodiment of a lateral row anchor repair using two anchors **80a**, **80b** disposed in bone **106** in a lateral row, each anchor **80a**, **80b** having a single construct **10a**, **10b** associated therewith, and two additional constructs **10c**, **10d** extending from the medial or primary fixation row. In the illustrated embodiment the third and fourth constructs **10c** and **10d** are double-loaded onto an anchor **80c**, although in other embodiments each construct can be associated with its own anchor, or the constructs **10c**, **10d** can be coupled to tissue without use of an anchor using techniques described herein or otherwise known to those skilled in the art. The

anchors **80a**, **80b**, **80c** are configured like the anchor **80** of FIG. 2, and the constructs **10a**, **10b**, **10c**, and **10d** are configured like the construct **10** of FIG. 2, but the addition of the letters “a,” “b,” “c,” and “d” are included for identification purposes of the separate implants.

(63) As shown, the anchors **80a**, **80b** are disposed in the bone **106** in a lateral row, with the constructs **10a**, **10b** each having a loop and a tail as described herein. The loop for each construct **10a**, **10b** can be configured to receive portions of the third or fourth constructs **10c**, **10d**, as well as the tail of its own construct **10a**, **10b**. For example, a Lark's Head cinch loop can form a receiving opening **52a**, **52b** in the loop to receive the third or fourth constructs **10c**, **10d** and the respective construct tail and collapse around them. Alternatively, the loop can have another collapsible configuration that is capable of collapsing around a construct **10c**, **10d** and the tail of the respective construct **10a**, **10b** of the loop.

(64) In the illustrated embodiment, the loop and tail of the construct **10c** can be passed through to a top side of the tissue **105** at locations M.sub.2 and N.sub.2 as shown. Likewise, the loop and tail of the construct **10d** can be passed through to a top side of the tissue **105** at locations P.sub.2 and Q.sub.2 as shown. The tail of the respective construct **10c**, **10d** can then be passed through its own receiving opening **52c**, **52d** formed in its loop, the receiving opening **52c**, **52d** can be collapsed, and the collapsed receiving opening **52c**, **52d** can be advanced toward the tissue **105** using techniques described herein. The coaxial region of the tails of the constructs **10c**, **10d** can be disassembled into limbs **56c**, **58c** and **56d**, **58d**. Optionally, one or more half-hitches can be formed using the limbs **56c**, **58c** and **56d**, **58d**. Regardless of whether any half-hitches are formed, the limbs **56c**, **58c**, and **56d**, **58d** can be extended towards the lateral row anchors **80a**, **80b**. In the illustrated embodiment the limbs **56c**, **58c**, and **56d**, **58d** are extended in a crossing manner such that the limbs **56c**, **58c** extend to the anchor **80a** that is on the opposite side of the medial anchor **80c**, and the limbs **56d**, **58d** extend to the anchor **80b** that is also on the opposite side of the medial anchor **80d**. A person skilled in the art will recognize that other configurations for associating the limbs **56c**, **58c** and **56d**, **58d** with the anchors **80a**, **80b** can also be used.

(65) As shown, the receiving opening **52a** associated with the anchor **80a** can receive the limbs **56c**, **58c**, as well as the tail of the construct **10a**, and then the receiving opening **52a** can be collapsed or dressed to capture the limbs **56c**, **58c** and the tail. Likewise, the receiving opening **52b** associated with the anchor **80b** can receive the limbs **56d**, **58d**, as well as the tail of the construct **10b**, and then the receiving opening **52b** can be collapsed or dressed to capture the limbs **56d**, **58d** and the tail. A pulling force can be applied to the respective tails, which in turn can pull the collapsed receiving openings **52a**, **52b** onto the respective anchors **80a**, **80b**. The coaxial region of the tails of the constructs **10a**, **10b** can be disassembled and used to form one or more half-hitches. While the half-hitches can set the location of the receiving openings **52a**, **52b** with respect to the anchors **80a**, **80b**, the limbs **56c**, **58c** and **56d**, **58d** of the constructs **10c**, **10d** can still be slid through the respective openings **52a**, **52b**. The limbs **56c**, **58c** and **56d**, **58d** can be used to form one or more half-hitches to set the location of the limbs **56c**, **58c** and **56d**, **58d** to prevent them from sliding.

(66) FIG. 11 illustrates one exemplary embodiment of using the construct **10** and anchor **80** of FIG. 2 in a tissue repair procedure. As shown, the anchor **80** having the construct **10** associated therewith is implanted in bone **106** in a medial or primary fixation row. The tail **16** of the construct **10** passes through the tissue **105** to a top side of the tissue at a location M.sub.3 and the loop **12** of the construct **10** passes through the tissue **105** to a top side of the tissue at a location N.sub.3. The tail **16** is passed through a portion of the loop **12**, and is then extended to an anchor **80'''** disposed in the lateral row in the bone **106**. Tension is applied to the tail **16**, and thus the rest of the construct **10**, by trapping the tail **16** between the anchor **80'''** and the bone **106**. As a result, the formation of a Lark's Head cinch loop and/or half-hitches can be optional.

(67) FIG. 12 illustrates another exemplary embodiment of using the construct **10** and anchor **80** of FIG. 2 in a tissue repair procedure. In this embodiment, a thicker footprint can be achieved by

using a thicker filament **20**, such as a #5 Ethibond™ filament. As shown, the anchor **80** having the construct **10** associated therewith can be implanted in a medial or primary fixation row. The loop and tail of the construct **10** can be passed through tissue **105** to a top side of the tissue at locations M.sub.4 and N.sub.4, and the tail can then be passed through the receiving opening **52** formed by the loop, such as a Lark's Head cinch loop. The receiving opening **52** can be collapsed around the tail, and then a coaxial region of the construct **10** can be deconstructed as described herein. The limbs **56**, **58** that result from the deconstruction of the coaxial region can extend side-by-side to an anchor in the lateral row for subsequent attachment. Because the filament **20** is of a thicker variety, the configuration of the construct **10** can be of a ribbon-like footprint.

(68) FIG. **13** illustrates one exemplary embodiment of using the anchor **80'** in conjunction with the three constructs **10**, **10'**, and **10''** illustrated in FIG. **3B**. In this embodiment, thinner filaments **20** can be used to more easily allow for all three constructs **10**, **10'**, **10''** to be operated without tangling or other undesirable obstructions. As shown, the tails can be passed through tissue **105** to a top side of the tissue at locations M.sub.5, N.sub.5, and P.sub.5, to engage with the loops associated with the anchor **80''**. The tails can then be passed through a receiving opening **52**, **52'**, **52''** formed by the loops, such as Lark's Head cinch loops. The receiving openings **52**, **52'**, **52''** can then be collapsed around the respective tails. The coaxial region can be deconstructed and then the limbs resulting therefrom can be used to tie one or more half-hitches to maintain a location of the collapsed receiving openings **52**, **52'**, **52''** using techniques described herein or otherwise known to those skilled in the art. In some embodiments, a rip stitch **90** can be disposed laterally across a portion of the tissue **105** to receive the collapsed receiving openings **52**, **52'**, **52''**. The rip stitch **90** can prevent undesirable tearing due to stress caused by the collapsed receiving openings **52**, **52'**, **52''**.

(69) FIG. **14** illustrates one exemplary embodiment for repairing a tear **109** in tissue **105**. In the illustrated embodiment, the construct **10**, **10'** is used in two different ways to repair the tear **109**. In one instance, the construct **10** is used by itself, without any attachment to any anchor or other implant device. The tail **16** and the loop **12** can extend from opposite sides of the tear **109** at locations M.sub.6 and N.sub.6 as shown. Then the construct **10** can be operated in a manner similar to as described herein for other embodiments. For example, a receiving opening can be formed from the loop **12**, the tail **16** can be disposed in the receiving opening, and the receiving opening can be collapsed or dressed around the tail **16**. Tension can then be applied to the tail **16** to advance the collapsed receiving opening against the tissue **105**, and thus adjacent to the tear **109**. As the receiving opening is collapsed, the portions of tissue **106** through which the tail **16** and loop **12** are disposed can be drawn towards each other.

(70) In the other instance, the construct **10'** is used in conjunction with an anchor (not shown) disposed in bone **106**. The tail and the loop extend from opposite sides of the tear **109** at locations P.sub.6 and Q.sub.6 as shown. Then the construct **10'** is operated as described with respect to the first instance in FIG. **14**. The illustrated embodiment shows the collapsed receiving opening **52'** with the tail disposed therethrough adjacent to the tear **109** and one or more half-hitches formed therein to lock the location of the collapsed receiving opening **52'** with respect to the tear **109**. Using the construct **10'** with the anchor provides for a secure attachment of the torn tissue **105** to bone **106** while also providing repair to the tear **109** itself. Use of the construct **10** without the anchor can be used to help repair the tear **109** even without attaching to bone, for example in instances in which the tear is located too far away from the bone to effectively use an anchor.

(71) FIG. **15** illustrates one exemplary embodiment of using the dual-loaded anchor **80'** of FIG. **3A** in an instability repair involving a glenoid or hip **106'** and labral tissue **107**. As shown, the anchor **80'** is implanted in the hip **106'** using techniques known to those skilled in the art. Each of the two tails **16**, **16'** are then passed from one side of the labral tissue **107** to the other side. The loops **12**, **12'** remain disposed on the same side of the labral tissue as the anchor **80'**. The tails **16**, **16'** and loops **12**, **12'** are then operated using techniques described herein or otherwise known to those skilled in the art. For example, a collapsible receiving opening can be formed in the loop **12**, **12'**

and the receiving opening can receive the tail **16**, **16'**. The tail **16**, **16'** can then be operated to advance the collapsed receiving opening toward the anchor **80'**, which in turn draws the labral tissue **107** toward the hip **106'**. The resulting collapsed receiving opening can be held in place by some combination of a Lark's Head cinching loop (if used to form the receiving opening for instance), one or more half-hitches, or extending limbs from a disassembled coaxial region toward another anchor to apply tension to the limbs.

(72) One skilled in the art will appreciate further features and advantages of the invention based on the above-described embodiments. Accordingly, the invention is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. A person skilled in the art would be able to adapt the constructs, anchors, and techniques provided for herein for suitable use with other combinations of constructs, anchors, and techniques without departing from the spirit of the present disclosure. Further, although the constructs and methods provided for herein are generally directed to surgical techniques, at least some of the constructs and methods can be used in applications outside of the surgical field. All publications and references cited herein are expressly incorporated herein by reference in their entirety.

## Claims

1. A surgical repair method, comprising: passing a terminal end of a surgical filament through soft tissue at a location proximate to bone in which the soft tissue is to be attached, the surgical filament having a first portion with a loop formed therefrom, the loop including a snare formed from a portion thereof, a second portion that includes the terminal end, and a coaxial region formed between the first and second portions by disposing a portion of the filament into its own volume, the terminal end and the loop extending proximally from the soft tissue; passing the terminal end of the second portion through the snare; collapsing the snare to engage the soft tissue; deconstructing the coaxial region by removing the portion of the filament from the volume of the filament in which it was disposed; associating the surgical filament with an anchor disposed in the bone; and advancing the collapsed snare distally to bring the tissue into proximity with the bone.
2. The method of claim 1, further comprising forming the snare from the loop of the first portion of the filament after passing the terminal end of a surgical filament through soft tissue.
3. The method of claim 1, further comprising inserting the anchor in bone in proximity to the soft tissue.
4. The method of claim 1, wherein passing the terminal end through soft tissue comprises passing the terminal end through the soft tissue from a proximal side to a distal side and then back through the soft tissue again to the proximal side.
5. The method of claim 1, wherein passing the terminal end through soft tissue comprises separately passing the terminal end and the loop through the soft tissue from a distal side to a proximal side.
6. The method of claim 1, wherein deconstructing comprises removing a second terminal end of the filament out of the volume out from the volume of the filament in which it was disposed such that the second terminal end extends proximally from the soft tissue.
7. The method of claim 1, wherein advancing the collapsed snare distally to bring the tissue into proximity with the bone comprises trapping the surgical filament between the anchor and bone and inserting the anchor into bone, the inserting tensioning the surgical filament to advance the collapsed snare distally.
8. The method of claim 1, wherein the anchor is a first anchor, the method further comprising, after associating the surgical filament with the first anchor disposed in bone, associating the surgical filament with a second anchor disposed in bone at a location proximate to the first anchor.
9. The method of claim 8, further comprising, associating the surgical filament with a second anchor and disposing the second anchor in bone at a location proximate to the first anchor, wherein

disposing the second anchor in bone comprises trapping the surgical filament between the second anchor and bone.

10. The method of claim 9, wherein trapping the surgical filament between the second anchor and bone comprises tensioning the surgical filament and further collapsing the snare.

11. The method of claim 1, further comprising tying at least one locking knot with the filament at a location that is proximate to the collapsed snare to secure a location of the filament with respect to the tissue.

12. A surgical repair method, comprising: passing a first terminal end of a surgical filament through soft tissue at a location proximate to bone in which the soft tissue is to be attached, the surgical filament having a first portion that includes the first terminal end, a second portion that includes a loop formed therefrom and a second terminal end, and a coaxial region formed between the first and second portions, by disposing the second terminal end of the surgical filament into its own volume, the first terminal end and the loop extending proximally from the soft tissue, and the loop of the second portion arranged to form cinch loop defining a receiving opening; passing the first terminal end of the surgical construct through the receiving opening; collapsing the receiving opening of the cinch loop; deconstructing the coaxial region by removing the second terminal end of the surgical filament from the volume of the surgical filament in which it was disposed; and associating the surgical filament with an anchor disposed in the bone; and advancing the collapsed cinch loop distally to bring the tissue into proximity with the bone.

13. The method of claim 12, further comprising forming the cinch loop in the second terminal end of the first surgical filament after the first terminal end has been passed through the soft tissue.

14. The method of claim 12, further comprising tying at least one locking knot with the filament at a location that is proximate to the collapsed snare to secure a location of the filament with respect to the tissue.

15. The method of claim 12, further comprising inserting the anchor in bone in proximity to the soft tissue.

16. The method of claim 12, wherein passing the first terminal end through soft tissue comprises passing the first terminal end through the soft tissue from a proximal side to a distal side and then back through the soft tissue again to the proximal side.

17. The method of claim 12, wherein passing the first terminal end through soft tissue comprises separately passing the first terminal end and the loop through the soft tissue from a distal side to a proximal side.

18. The method of claim 12, wherein advancing the collapsed cinch loop distally to bring the tissue into proximity with the bone comprises trapping the surgical filament between the anchor and bone and inserting the anchor into bone, the inserting tensioning the surgical filament to advance the collapsed cinch loop distally.

19. The method of claim 12, wherein the anchor is a first anchor, the method further comprising, after associating the surgical filament with the first anchor disposed in bone, associating the surgical filament with a second anchor disposed in bone at a location proximate to the first anchor.

20. The method of claim 19, further comprising, associating the surgical filament with a second anchor and disposing the second anchor in bone at a location proximate to the first anchor, wherein disposing the second anchor in bone comprises trapping the surgical filament between the second anchor and bone.

21. The method of claim 20, wherein trapping the surgical filament between the second anchor and bone comprises tensioning the surgical filament and further collapsing the snare.

22. The method of claim 12, further comprising: removing the second terminal end of the surgical filament from the coaxial region of the filament after collapsing the receiving opening of the cinch loop; and tying at least one knot with the filament at a location that is proximate to the collapsed cinch loop to secure a location of the filament with respect to the tissue.

23. The method of claim 12, wherein the surgical filament is associated with the anchor disposed in the bone after passing the first terminal end through the receiving opening.

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